

Deep Learning for Explainable Cancer Identification Spectroscopy

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2025

Abstract

This is some stuff

1 Introduction

Cancer is one of the leading causes of death worldwide with around 20 million cases reported in 2022. This corresponds to one in five men or women developing cancer in their lifetime [1]. The conventional method for cancer diagnosis, as outlined in Figure 1, involves taking a biopsy of suspect tissue from a patient during surgery, fixing this biopsy to a slide, and then staining the biopsy with haematoxylin and eosin (H&E). The haematoxylin stains nucleic acids a deep blue-purple colour, whilst the eosin is pink and non-specifically stains proteins [2]. These H&E stained samples can then be analysed by a pathologist to develop a diagnosis. This process is manual, error-prone, subjective, costly, and time-consuming; often requiring transport to a laboratory for processing by highly trained technicians [3]. The sample processing time alone is often 20 minutes or more [4], so the development of methods which forgo traditional lab-based preparation have significant potential to guide surgeries more effectively.

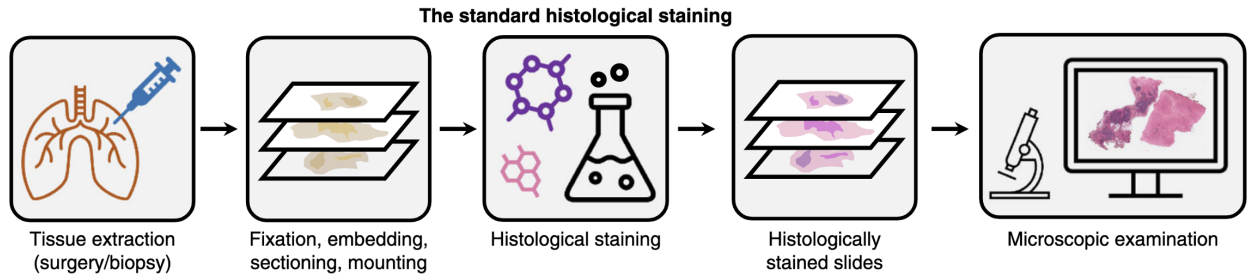


Figure 1: From [5]

New methods are being developed to expedite the cancer diagnosis process by using machine learning in conjunction with a range of imaging technologies including stimulated Raman spectroscopy (SRS) [3, 6–8], second harmonic generation microscopy (SHG) [6], and Fourier transform infrared spectroscopy (FTIR) [9, 10]. Within this work, I focus on the application of FTIR spectroscopy to cancer diagnosis within colon cancer samples provided by collaborators at the University of Exeter, which were collected and had FTIR spectral images produced as part of the MINERVA project [11].

1.1 Previous Works

Work by Tomas et al. [9] utilised attenuated total reflection Fourier transform infrared (ATR-FTIR) spectroscopy and feed-forward full-connected neural networks to classify breast cancer samples into benign and malignant classifications with $96.06\% \pm 7.07\%$ accuracy, corresponding to the large differences in spectral characteristics between classes as can be seen in figure 2.

1.2 Future Vision

The TROPHY project [?] is aiming to develop a new vibrational spectroscopic imaging microscope exploiting photo-thermal infrared (PTIR), FTIR and digital holography microscopy (DMH) to evaluate cell-type

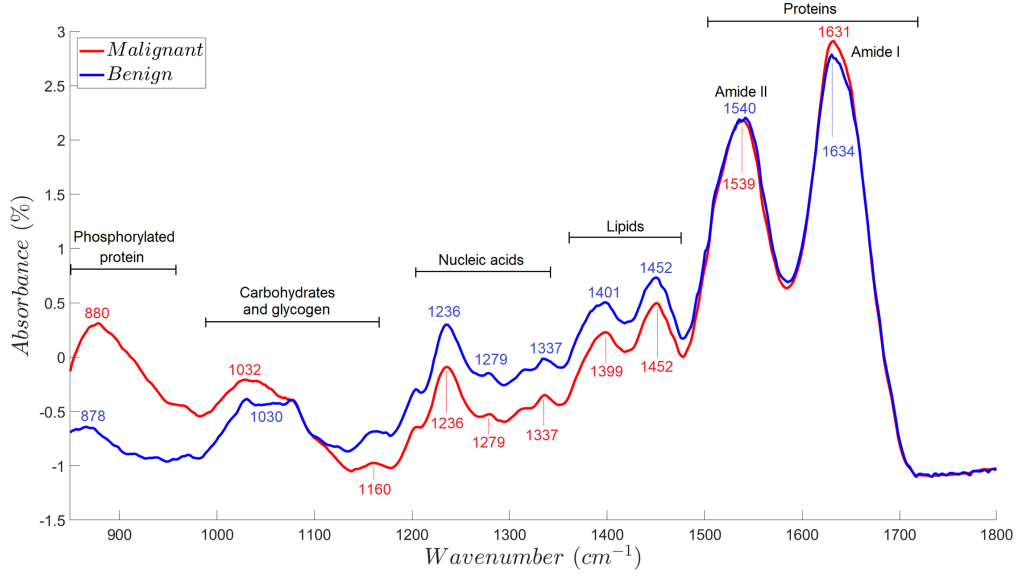


Figure 2: From [9]

topologies, identify mutant cell clusters, and provide biochemical analyses to determine tumour types, grading and staging [?]. The MINERVA and TROPHY datasets are expected to be similar, so the models and methods developed in this work constitute a basis for future work towards the TROPHY goals. This is particularly apparent given that TROPHY will focus primarily on the same fingerprint 1000-1800 cm^{-1} region of the IR spectrum as MINERVA.

Other works which could benefit from the approaches described here include work on the CHARM project which is also taking place in Cambridge. This project is similar to TORPHY in that they are developing a new microscope for cancer diagnosis. CHARM are instead focussing on Raman, SGH and TPEF signals, as these have been shown to have some advantages over FTIR [?]. Currently, data produced by that microscope is very noisy, and also only has intensity information at around 20 wavelengths, limiting the viability of the SCNN method implemented in this work. However, as the microscopy performance improves, it should be possible to apply the same methods as described here. Some work was done throughout this project to remove vignetting effects in the CHARM dataset. However, this was only pursued briefly, so information relating to it is attached in appendix ??

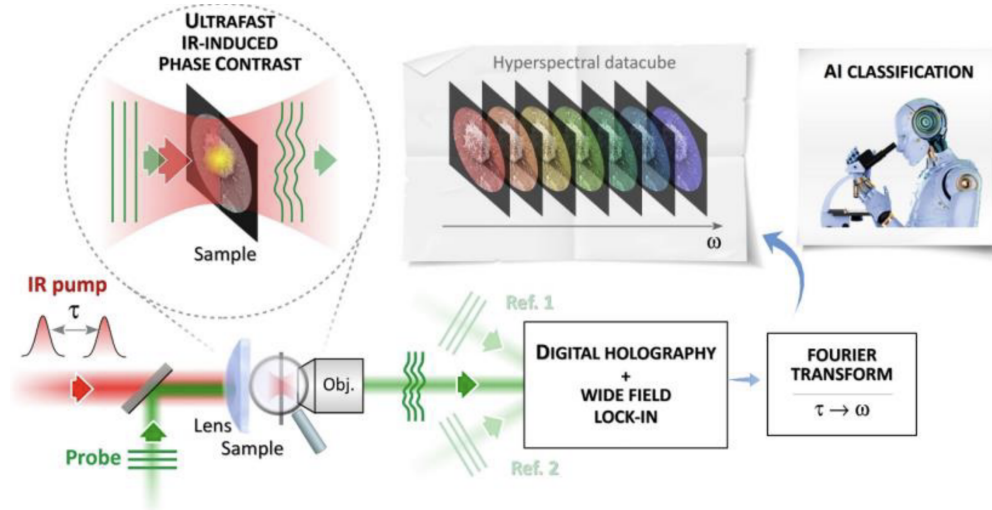


Figure 3: From Trophy proposal [12]

1.3 Novelty of This Work

This work is focusses on developing a deep learning method for cancer identification from FTIR data which is both explainable and generalisable. This is achieved by first implementing a physically-motivated convolutional neural network, before then utilising a context-representative method to create a latent space from within which different clusters can be identified to elucidate the multiple pathways by which an individual classification such as cancer or non-cancer can be achieved. No other work has been found to use a similar method, although others have used the context-representative method to extract different spectral contributions within urine samples [?]. This work then further extends this method, by utilising the CRIME predictions of average spectra spectra and weightings describing the importance of each wavelength within those spectra, to then produce a simple weighted cosine similarity model for classification into non-cancer, hyperplastic, adenoma and cancerous classes.

2 Methods

2.1 FTIR Spectroscopy

Infrared spectroscopy generally works by directing a broadband spectrum with wavenumbers ranging from around $400 - 4000 \text{ cm}^{-1}$ (for mid-IR spectroscopy) [13] onto a sample. This coherent spectrum then excites resonant vibrational modes within molecular bonds inside the sample, causing absorption of specific frequencies depending on the composition of the incident sample. This decreases the intensity relative to the baseline in the absorbed frequencies, therefore allowing a characteristic spectrum for each bond to be established. Modern devices use a Fourier transform-based (FTIR) interferogram approach [14], enabling high signal-to-noise (SNR) ratios to be achieved.

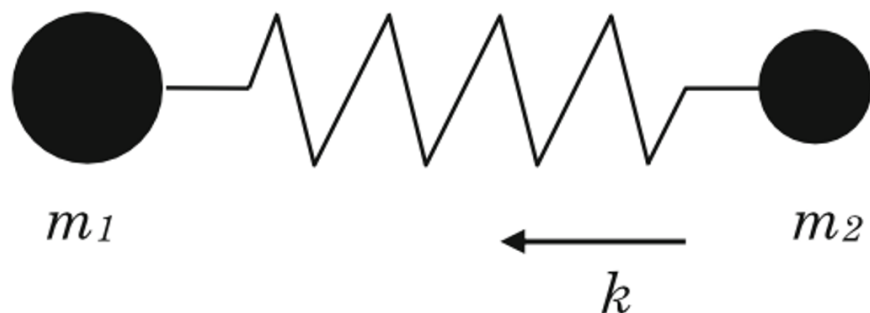
FTIR can detect biochemical compositions including nucleic acids, proteins, lipids and carbohydrates within biological samples by matching the observed spectra with known forms for different bonding types, functional groups and intermolecular interactions [13]. Typical absorption groups within biological samples are detailed in Table 1.

Wavenumber(cm^{-1})	Assignment
3080–2800	Anti-symmetric and symmetric C–H stretches from proteins and lipids
1745–1725	Ester carbonyl of lipids
1700–1500	Amide I and II groups in peptide linkages of proteins
1270–1080	Anti-symmetric and symmetric C–O and P–O areas in DNA, RNA and phospholipids
1200–900	Carbohydrate vibrations of glucose, fructose and glycogen

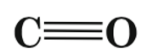
Table 1: Assignments of Various Functional Groups Based on Wavenumber

2.1.1 FTIR Dataset Details

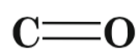
The MINERVA dataset used in this work contains FTIR spectral images of 90 colon cancer tissue samples. These images were produced by collaborators at the University of Exeter using an Agilent 620 FTIR microscope, an Agilent 670 FTIR spectrometer with a Global light source, and a liquid-nitrogen cooled 128x128 FPA detector [17]



Effect of chemical bonds :



$\sim 2100 \text{ cm}^{-1}$

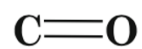


$\sim 1680\text{-}1630 \text{ cm}^{-1}$



$\sim 1200 \text{ cm}^{-1}$

Effect of H-bonding :

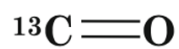


$\sim 1680\text{-}1630 \text{ cm}^{-1}$

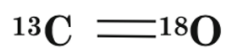


$\Delta\nu \sim - (10\text{-}20) \text{ cm}^{-1}$

Effect of mass :



$\Delta\nu \sim - (35\text{-}55) \text{ cm}^{-1}$



$\Delta\nu \sim - 65 \text{ cm}^{-1}$

Figure 4: From [15]

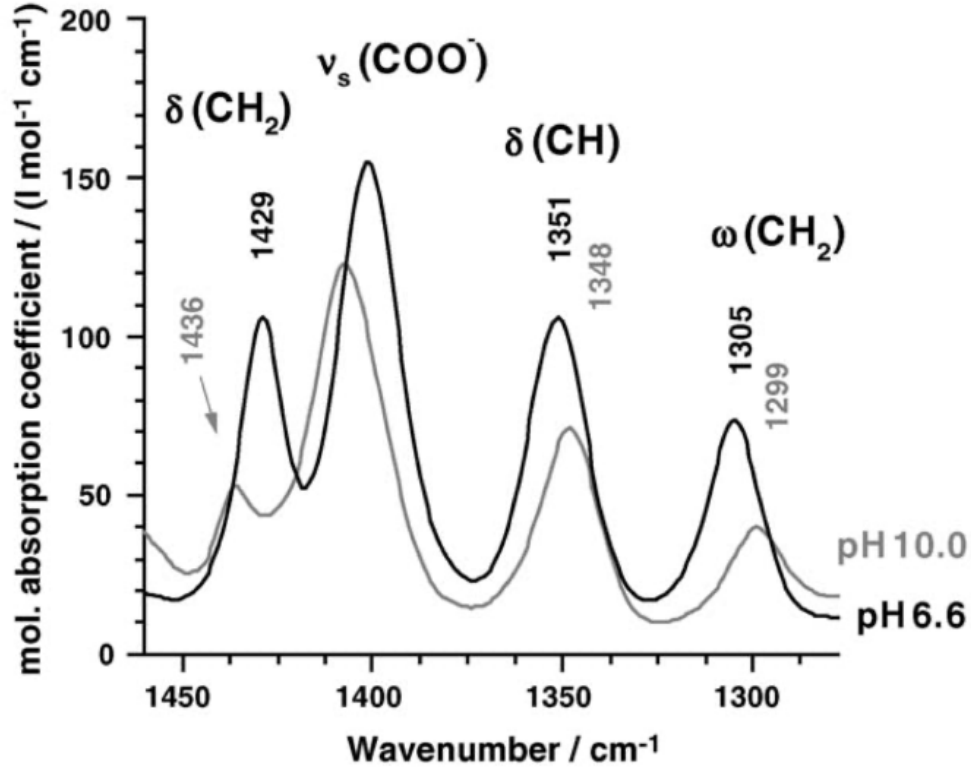


Figure 5: From [16]

2.2 Machine Learning

3 Results

3.1 Dataset Analysis

The data for this work was collected as described in Section ??, resulting in a dataset containing 90 colon biopsy samples, their associated H&E stained images, their FTIR spectral images with absorption intensities at 801 wavenumbers from 1000-1800 cm^{-1} , and their histopathological classification as either normal, hyperplastic, adenomatous or cancerous tissue [17]. Notably, this histopathological classification was made according to the most severe tissue condition present within the sample, so a sample containing only 2-3% cancerous tissue would still be classified as cancerous. We therefore expect many individual pixels within the spectral images to be incorrectly labelled.

False-colour images for samples were generated by taking the intensities for each pixel at high-variance wavenumbers 1080 cm^{-1} , 1230 cm^{-1} and 1600 cm^{-1} , and then scaling them to values between 0 and 1, centred around 0.5 according to the standard deviation of intensity values at that wavenumber throughout the sample. For each classification, one of these false-colour images along with its associated H&E stain is presented in Figure 6. This figure both demonstrates the boundary of stroma regions and the locations of H&E image folds preventing perfect alignment. Only the non-stroma regions are used for subsequent prediction analysis. Furthermore, comparisons between the intensities and variances with wavelength for each classification are shown in Figure 7.

3.2 Spectral Matching Methods

The most intuitive approach for classifying spectra according to their severity is to find reference spectra for each classification and then compute which reference spectra most closely resembles a test spectrum. Similarly to work by Villazon et al [18], this work implemented a mean square error (MSE) method, a cosine

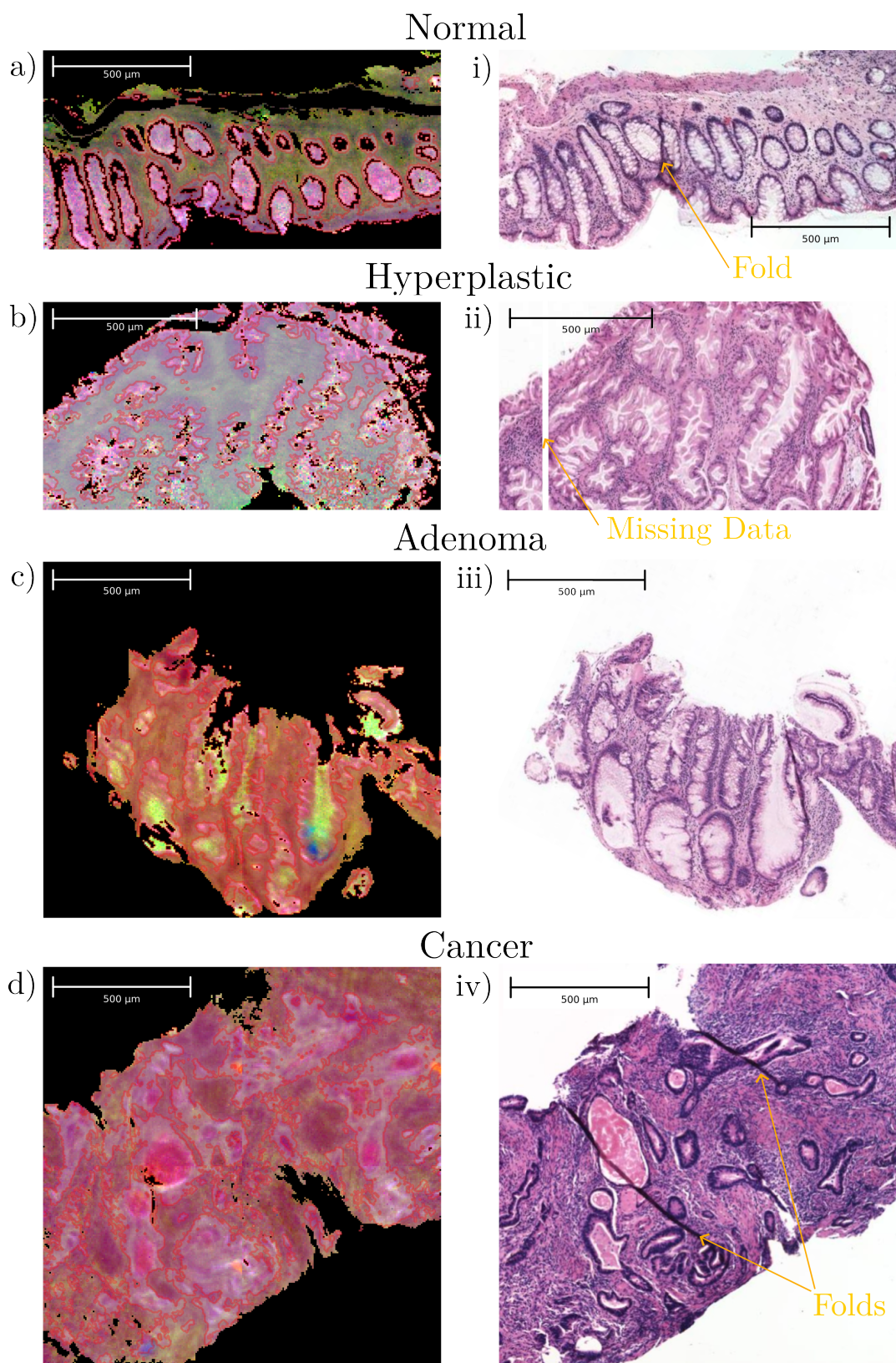


Figure 6: Sections a-d show false-colour FTIR images of colon biopsy samples - one per labelling classification. Each of these has non-stroma regions surrounded by red outlines and highlighted with a white accent. Sections i-iv show the corresponding aligned H&E images. Note that the H&E images in sections i and iv show folding defects, and information for a small stripe of the image in section ii is missing, so is blanked.

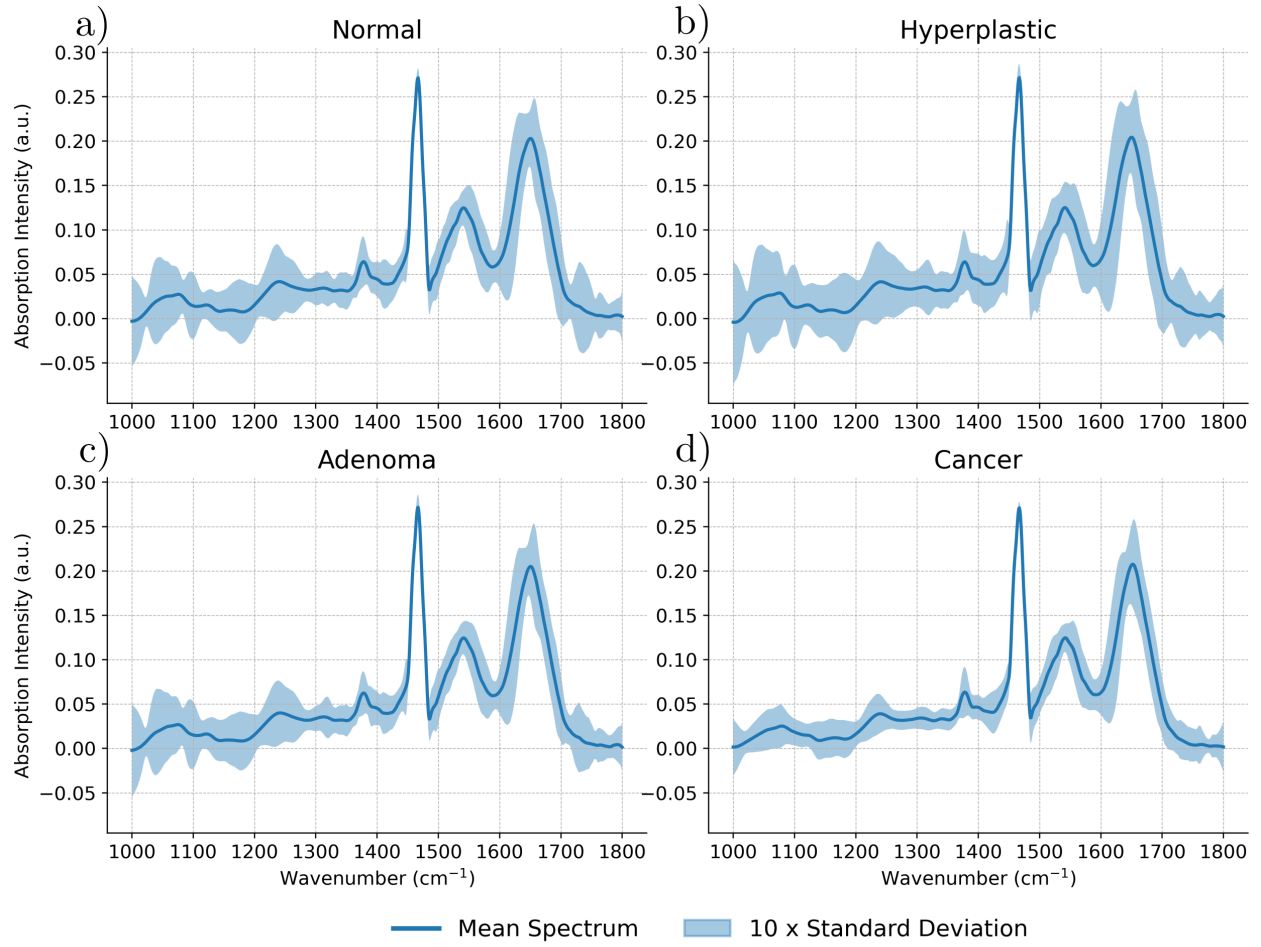


Figure 7: Plots showing the mean and standard deviation of the FTIR spectral intensities across each class.

similarity method. The mean square error is defined as:

$$\text{MSE}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2 \quad (1)$$

where x and y are the two spectra being compared, and n is the number of wavenumbers in each spectrum. The cosine similarity is defined as:

$$\text{Cosine Similarity}(x, y) = \frac{x \cdot y}{\|x\| \|y\|}. \quad (2)$$

By then adding 1 to the cosine similarity and dividing by 2, it is scaled to lie between 0 and 1 with 0 denoting complete orthogonality and 1 denoting complete similarity.

3.3 Traditional Machine Learning Approaches

3.4 A Scanning Convolutional Neural Network Approach

3.5 LIME Explanations

3.6 CRIME Explanations

3.6.1 VAE Method

3.6.2 M-VAE Method

3.7 CRIME Predictions

4 Discussion

5 Conclusions

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